

A Spatial Theory of Exchange Rate Disconnect

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Abstract

This paper studies how geographic distance shapes exchange-rate disconnect and international risk sharing. I first document that the Backus–Smith correlation declines with bilateral distance, relative consumption becomes more closely tied to relative output among distant country pairs, and the Fama relationship deteriorates with distance. To interpret these patterns, I develop a two-country open-economy model with financial demand shocks and geographic information frictions. Arbitrageurs observe foreign asset returns imperfectly, with information risk increasing in bilateral distance. Greater distance therefore reduces arbitrage capacity and amplifies the uncovered-interest-parity wedge generated by a given financial shock. The model also incorporates distance-dependent trade costs, which govern the spatial transmission of real shocks. Quantitatively, the model reproduces the decline in the Backus–Smith correlation, the increase in exchange-rate volatility relative to consumption and output, larger UIP deviations, and a more negative Fama coefficient as distance rises. These results suggest that international financial integration and exchange-rate disconnect vary systematically across geographic space.

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1 Introduction

Exchange rates are among the most volatile asset prices in the global economy, yet their movements remain difficult to reconcile with standard macroeconomic fundamentals. Two prominent manifestations of this disconnect are the Backus–Smith puzzle, which refers to the weak relationship between relative consumption and real exchange rates (Backus and Smith, 1993), and the forward premium puzzle, which reflects persistent departures from uncovered interest parity (Fama, 1984). A growing literature emphasizes financial shocks and imperfect international financial intermediation as important sources of these patterns (Gabaix and Maggiori, 2015; Maggiori, 2017; Itskhoki and Mukhin, 2021). In particular, financial demand shocks can generate large exchange-rate movements even in the absence of corresponding changes in real fundamentals.

At the same time, international financial integration has a pronounced geographic structure. Cross-border financial transactions and asset holdings decline with bilateral distance, a pattern commonly associated with information frictions, limited familiarity, and other barriers to investing in distant markets (Portes et al., 2001; Portes and Rey, 2005; Aviat and Coeurdacier, 2007; Okawa and van Wincoop, 2012). Geographic structure is also present in exchange rates and international business-cycle comovement (Lustig and Richmond, 2020; Servén and Abate, 2020). These observations raise a natural question: if international financial intermediation itself is geographically segmented, does the strength of exchange-rate disconnect also vary systematically across space?

This paper studies the geographic dimension of exchange-rate disconnect and international risk sharing. I begin by documenting three empirical patterns. First, the Backus–Smith correlation declines with bilateral geographic distance. Country pairs located farther apart exhibit a weaker relationship between relative consumption growth and changes in the real exchange rate. Second, international consumption risk sharing deteriorates with distance. Relative consumption responds more strongly to relative output fluctuations among geographically distant country pairs, and pair-specific measures of risk sharing decline systematically with bilateral distance. Third, the Fama relationship also exhibits a geographic gradient. When the standard Fama regression is augmented with an interaction between the interest-rate differential and bilateral distance, the implied Fama coefficient declines as countries become farther apart. Taken together, these results suggest that geography is associated not with a single exchange-rate moment, but with a broader deterioration in

international financial integration.

To interpret these patterns, I develop a two-country open-economy model with financial demand shocks and geographic information frictions. The model builds most directly on the financial-shock mechanism of Itskhoki and Mukhin (2021). Noise traders generate shifts in international asset demand, while arbitrageurs take positions against these disturbances. The key departure from the standard environment is that arbitrageurs observe foreign asset returns imperfectly, and the variance of this information noise increases with bilateral geographic distance. Distance therefore affects international financial integration through the risk associated with evaluating and intermediating foreign positions.

This friction generates a simple mechanism linking geography to exchange-rate disconnect. Greater information risk makes arbitrageurs less willing to absorb a given financial demand shock. A larger expected return differential is therefore required to induce the arbitrage position necessary to clear international asset markets. As a result, the uncovered-interest-parity wedge generated by a given financial shock increases with geographic distance. Formally, if ϕ denotes log bilateral distance and $\chi_1(\phi)$ is the loading of the UIP wedge on the financial shock, the model implies

$$\frac{\partial \chi_1(\phi)}{\partial \phi} > 0.$$

Thus, identical financial shocks can generate systematically different exchange-rate responses depending on the geographic distance between the countries involved. Geography matters not simply because distant countries receive different shocks, but because the international financial system transmits a given shock differently across space.

Financial information frictions alone, however, need not generate all of the geographic patterns observed in real-side risk sharing. I therefore also allow iceberg trade costs to increase with bilateral distance, following the view that trade frictions play an important role in international macroeconomic puzzles (Obstfeld and Rogoff, 2000). The two geographic channels operate through distinct margins. Information frictions govern how financial shocks are absorbed through international asset markets, while trade costs shape the transmission of real shocks across countries. Productivity and financial shocks consequently have different implications for consumption–output comovement, and their interaction with these two geographic frictions allows the model to speak jointly to the spatial behavior of exchange rates, consumption, and output.

I evaluate the model quantitatively by varying bilateral distance from 500 to 12,000 kilometers while holding the remaining structural parameters fixed. The model reproduces several of the geographic gradients observed in the data. The Backus–Smith correlation falls from 0.038 to -0.018 , compared with a decline from approximately 0.03 to -0.06 in the data. The geographic slope of exchange-rate volatility relative to consumption is 0.26 in the model and 0.23 in the data, while the corresponding slopes relative to output are 0.25 and 0.28. The standard deviation of UIP deviations rises by approximately 42 percent between 500 and 12,000 kilometers, compared with an increase of roughly 50 percent in the data. The model also generates a more negative Fama coefficient as bilateral distance increases. Although the model does not match the level of every moment exactly, it reproduces the common geographic movement across several measures of exchange-rate disconnect and international risk sharing.

The paper makes three contributions. First, it documents a systematic geographic dimension of exchange-rate disconnect. The Backus–Smith relationship, consumption risk sharing, and the Fama relationship all vary with bilateral distance. These results show that the degree of international financial integration is spatially heterogeneous across country pairs.

Second, the paper introduces geographic information frictions into a model of financially driven exchange-rate fluctuations. Distance does not enter merely as a reduced-form wedge in the UIP condition. Instead, it changes the information risk faced by international arbitrageurs and therefore their willingness to intermediate financial shocks. This mechanism delivers an analytical prediction linking bilateral distance to the magnitude of the UIP wedge generated by a given financial shock. Third, the paper shows quantitatively that this mechanism can account for several spatial features of international macroeconomic data within a common framework. Together with distance-dependent trade costs, geographic information frictions generate systematic variation in the Backus–Smith correlation, relative exchange-rate volatility, UIP deviations, and the Fama coefficient across bilateral distance.

Related literature. The paper relates to several strands of the literature. Most directly, it contributes to the literature explaining exchange-rate disconnect through imperfect financial intermediation and financial shocks. Gabaix and Maggiori (2015) emphasize the role of limited financial intermediation in exchange-rate determination, while Maggiori (2017) studies the implications of

financial intermediation for international risk sharing. Itskhoki and Mukhin (2021) show that financial shocks can jointly account for several major exchange-rate puzzles in general equilibrium, and Itskhoki and Mukhin (2025) further study the role of financial forces in exchange-rate dynamics. I build on this literature by allowing the strength of international financial intermediation to vary systematically with bilateral geographic distance.

The paper is also related to the literature documenting gravity in international finance. Portes et al. (2001) and Portes and Rey (2005) show that international financial transactions decline with geographic distance and emphasize information-related explanations for this pattern. Aviat and Coeurdacier (2007) document the joint geography of trade and asset holdings, while Okawa and van Wincoop (2012) develop a gravity framework for international asset allocation. These papers establish that geography affects the quantity and direction of international capital allocation. The mechanism in this paper instead emphasizes how geographic information frictions affect the transmission of financial shocks and, consequently, exchange-rate dynamics.

Finally, the paper relates to work studying spatial structure in international macroeconomic and financial outcomes. Lustig and Richmond (2020) document a gravity structure in exchange-rate factors, while Servén and Abate (2020) emphasize spatial dependence in international business cycles. The present paper complements this evidence by focusing on a specific mechanism through which geography affects exchange-rate disconnect: distance-dependent information risk limits international arbitrage and amplifies the effects of financial shocks across distant country pairs.

The remainder of the paper proceeds as follows. Section 2 presents the empirical evidence on the geographic structure of international risk sharing and exchange-rate disconnect. Section 3 develops the model and derives the role of geographic information frictions in the UIP wedge. Section 4 presents the quantitative analysis and compares the model-implied geographic gradients with their empirical counterparts. Section 5 concludes.

2 Motivating Empirical Evidence

This section documents a geographic dimension of international risk sharing and exchange-rate disconnect. If barriers to international financial integration increase with geographic distance, countries located farther apart should exhibit weaker international risk sharing and larger deviations

from standard international asset-pricing relationships. I examine this idea using three empirical exercises.

First, I ask whether the Backus–Smith relationship varies systematically with bilateral geographic distance. Second, I examine whether the sensitivity of relative consumption to relative output shocks increases with distance, as would be expected when risk sharing becomes more limited. Third, I study whether deviations from uncovered interest parity exhibit a similar geographic pattern using a modified Fama regression. Across these exercises, geographically distant country pairs display systematically weaker international risk sharing and a stronger exchange-rate disconnect.

2.1 The Backus–Smith Relationship and Geographic Distance

Under complete international financial markets, relative consumption should move closely with the real exchange rate. In particular, standard international risk-sharing conditions imply a strong positive relationship between changes in the real exchange rate and relative consumption growth. In the data, however, this relationship is typically weak and is often negative, a pattern commonly referred to as the Backus–Smith puzzle.

I examine whether the extent of this disconnect varies systematically with geographic distance. For each country pair (i, j) , I first compute the time-series correlation between changes in the bilateral real exchange rate and relative consumption growth,

$$BS_{ij} \equiv \text{corr}(\Delta q_{ij,t}, \Delta c_{i,t} - \Delta c_{j,t}).$$

I then estimate the cross-sectional relationship

$$BS_{ij} = \alpha + \beta \ln(\text{dist}_{ij}) + \varepsilon_{ij}.$$

A negative coefficient β indicates that the Backus–Smith relationship deteriorates with geographic distance: countries located farther apart exhibit a weaker comovement between relative consumption and the real exchange rate and, therefore, a larger departure from the complete-markets benchmark. Table ?? reports the results. The coefficient on log geographic distance is negative across specifications. In the baseline specification, the estimated coefficient is -0.018 , while the specification controlling for country fixed effects yields an estimate of -0.012 . Thus, geographically distant country pairs exhibit systematically lower Backus–Smith correlations.

This result suggests that the Backus–Smith puzzle itself has a spatial dimension. The failure of relative consumption to comove with the real exchange rate is not uniform across country pairs: the disconnect becomes more pronounced as bilateral geographic distance increases.

2.2 Risk Sharing and Geographic Distance

The spatial pattern in the Backus–Smith correlation suggests that international risk sharing may deteriorate with geographic distance. To examine this mechanism more directly, I study how strongly relative consumption responds to relative output shocks across country pairs.

Under effective international risk sharing, country-specific output fluctuations need not translate one-for-one into relative consumption fluctuations. In contrast, when financial markets are segmented, consumption becomes more closely tied to domestic output. I therefore estimate

$$\Delta c_{ij,t} = \alpha_{ij} + \tau_t + \beta_y \Delta y_{ij,t} + \gamma_y (\Delta y_{ij,t} \times \ln \text{dist}_{ij}) + \varepsilon_{ij,t},$$

where

$$\Delta c_{ij,t} \equiv \Delta c_{i,t} - \Delta c_{j,t}, \quad \Delta y_{ij,t} \equiv \Delta y_{i,t} - \Delta y_{j,t}.$$

The interaction coefficient γ_y captures how the transmission of relative output fluctuations into relative consumption varies with bilateral distance. A positive value of γ_y implies that relative consumption responds more strongly to relative output shocks for geographically distant country pairs, consistent with weaker international risk sharing.

Table ??, Panel A, reports the estimates. Without the distance interaction, the coefficient on relative output growth is 0.518. Once the interaction with geographic distance is introduced, the estimated interaction coefficient is positive and statistically significant, with $\hat{\gamma}_y = 0.078$. Thus, the sensitivity of relative consumption to relative output increases systematically with geographic distance.

As an alternative way to summarize this relationship, I estimate a pair-specific risk-sharing coefficient.

In the first stage, I estimate

$$\Delta c_{ij,t} = \alpha_{ij} + \tau_t + \beta_{ij}^{RS} \Delta y_{ij,t} + e_{ij,t}.$$

I then define the degree of risk sharing as

$$RS_{ij} \equiv 1 - \beta_{ij}^{RS}.$$

A larger value of RS_{ij} therefore corresponds to greater international consumption risk sharing. In the second stage, I relate this measure to geographic distance:

$$RS_{ij} = \beta_0 + \beta_1 \ln(\text{dist}_{ij}) + \nu_{ij}.$$

Panel B reports a negative and statistically significant relationship between risk sharing and bilateral distance. The estimated coefficient on log distance is -0.070 in the baseline specification and remains negative at -0.040 after including origin and destination fixed effects.

Taken together, the direct interaction and two-step specifications point to the same conclusion. Relative consumption is more closely tied to relative output among geographically distant countries, implying that international risk sharing declines with distance.

2.3 UIP Disconnect and Geographic Distance

I next examine whether the geographic pattern in international risk sharing is also present in the uncovered interest parity relationship. Under uncovered interest parity, an interest-rate differential between two countries should be offset by the expected change in their bilateral exchange rate. The standard Fama regression can be written as

$$\Delta e_{ij,t+1} = \beta_1 + \beta_2 (i_{i,t} - i_{j,t}) + \varepsilon_{ij,t+1},$$

where the UIP benchmark implies $\beta_2 = 1$.

To examine whether this relationship varies with geographic distance, I augment the Fama regression with an interaction between the interest-rate differential and bilateral distance:

$$\Delta e_{ij,t+1} = \beta_1 + \beta_2 (i_{i,t} - i_{j,t}) + \beta_3 [(i_{i,t} - i_{j,t}) \times \ln(\text{dist}_{ij})] + \alpha_{ij} + \tau_t + \varepsilon_{ij,t+1}.$$

The implied Fama coefficient for a country pair at distance d_{ij} is therefore

$$\beta^F(d_{ij}) = \beta_2 + \beta_3 \ln(d_{ij}).$$

A negative value of β_3 implies that the response of the exchange rate to an interest-rate differential becomes smaller as geographic distance increases.

Table ?? reports the results. In the interaction specification, the coefficient on the interest-rate differential is 1.786, while the coefficient on its interaction with log distance is -0.202 and statistically

significant. Hence, the implied Fama coefficient declines systematically with bilateral geographic distance.

This result provides a third spatial pattern. Countries that are farther apart not only exhibit weaker consumption risk sharing, but also display a systematically different exchange-rate response to interest-rate differentials. The geographic pattern in the Fama coefficient is consistent with financial-market segmentation becoming more important across distant country pairs.

2.4 Summary of the Empirical Evidence

The empirical results reveal a common geographic pattern across several measures of international financial integration. First, the Backus–Smith correlation declines with bilateral distance. Second, relative consumption responds more strongly to relative output shocks among distant country pairs, indicating weaker international risk sharing. Third, the Fama coefficient declines with geographic distance.

These findings suggest that geographic distance is associated with the strength of international financial integration rather than with a single exchange-rate moment alone. They motivate a model in which international financial intermediation is subject to bilateral geographic frictions. In the next section, I introduce such a friction through imperfect information about foreign asset returns and study how it affects the transmission of financial shocks to exchange rates and international risk sharing.

3 Model

This section develops a two-country open-economy model in which geographic distance affects international financial integration through imperfect information about foreign asset returns. The model builds on the financial-shock mechanism in Itskhoki and Mukhin (2021), while introducing a bilateral information friction whose magnitude increases with geographic distance.

There are two countries, Home and Foreign. Households consume domestic and foreign goods, supply labor, and trade international bonds. Firms produce differentiated goods using labor and trade is subject to iceberg costs. International financial markets contain noise traders, whose demand generates financial shocks, and arbitrageurs, who intermediate across countries. The key departure

from the standard environment is that arbitrageurs observe foreign asset returns with noise that increases with bilateral geographic distance.

This information friction changes the transmission of financial shocks across countries. For a given financial shock, greater bilateral distance reduces the arbitrage response and generates a larger deviation from uncovered interest parity. The resulting mechanism provides a link between geographic distance and the spatial patterns in international risk sharing documented in Section ??.

3.1 Households

The representative household in the Home country chooses consumption and labor to maximize expected lifetime utility,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{L_t^{1+\psi}}{1+\psi} \right], \quad (1)$$

subject to the nominal budget constraint

$$P_t C_t + \frac{B_{t+1}}{R_t} \leq W_t L_t + B_t + \Pi_t, \quad (2)$$

where C_t denotes aggregate consumption, L_t is labor supply, P_t is the consumer price index, W_t is the nominal wage, and Π_t denotes firm profits. The variable B_{t+1} denotes the household's bond position and R_t is the associated gross nominal interest rate.

The Foreign household faces an analogous problem. Foreign-country variables are denoted by an asterisk.

3.2 Consumption

Aggregate consumption is a CES composite of Home and Foreign goods,

$$C_t = \left[(1-\gamma)^{1/\theta} C_{H,t}^{(\theta-1)/\theta} + \gamma^{1/\theta} C_{F,t}^{(\theta-1)/\theta} \right]^{\theta/(\theta-1)}, \quad (3)$$

where γ governs the expenditure share on imported goods and θ is the elasticity of substitution between Home and Foreign goods.

Cost minimization implies demand for foreign goods,

$$C_{F,t} = \gamma \left(\frac{P_{F,t}}{P_t} \right)^{-\theta} C_t, \quad (4)$$

and the corresponding consumer price index is

$$P_t = \left[(1 - \gamma) P_{H,t}^{1-\theta} + \gamma P_{F,t}^{1-\theta} \right]^{1/(1-\theta)}. \quad (5)$$

The Foreign household has the symmetric consumption aggregator and demand system.

3.3 Production and Trade Costs

Firms in country $i \in \{H, F\}$ produce output using labor according to

$$Y_{i,t} = e^{a_{i,t}} L_{i,t}, \quad (6)$$

where productivity follows

$$a_{i,t} = \rho_a a_{i,t-1} + \sigma_a \varepsilon_{i,t}^a. \quad (7)$$

Under monopolistic competition, the producer price is a constant markup over marginal cost,

$$P_{i,t} = \frac{\theta}{\theta - 1} MC_{i,t} = \frac{\theta}{\theta - 1} e^{-a_{i,t}} W_{i,t}. \quad (8)$$

International trade is subject to iceberg trade costs. The price of the Foreign good faced by Home consumers is

$$P_{F,t} = \tau E_t P_{F,t}^*, \quad (9)$$

while the price of the Home good in the Foreign market is

$$P_{H,t}^* = \tau \frac{P_{H,t}}{E_t}. \quad (10)$$

We allow the bilateral iceberg trade cost to vary with geographic distance according to

$$\ln \tau_{ij} = \alpha_\tau + \eta_\tau \ln \text{dist}_{ij}. \quad (11)$$

The trade-cost channel is conceptually distinct from the financial information friction introduced below. Trade costs determine how real shocks and expenditure are transmitted across countries, whereas geographic information frictions operate through international asset markets and the transmission of financial shocks.

3.4 Market Clearing

Goods-market clearing in the Home country requires

$$Y_t = C_{H,t} + \tau C_{H,t}^*. \quad (12)$$

Using the consumption-demand equations, this condition can be written as

$$Y_t = (1 - \gamma) \left(\frac{P_{H,t}}{P_t} \right)^{-\theta} C_t + \tau \gamma \left(\frac{P_{H,t}^*}{P_t^*} \right)^{-\theta} C_t^*. \quad (13)$$

Home net exports are

$$NX_t = E_t P_{H,t}^* C_{H,t}^* - P_{F,t} C_{F,t}. \quad (14)$$

The evolution of the international bond position therefore satisfies

$$\frac{B_{t+1}}{R_t} - B_t = NX_t. \quad (15)$$

3.5 Financial Shocks

International asset markets are subject to demand shocks generated by noise traders. Noise-trader demand for the Foreign asset is given by

$$\frac{N_{t+1}^*}{R_t^*} = n \left(e^{\psi_t} - 1 \right), \quad (16)$$

where the financial demand shock follows

$$\psi_{ij,t} = \rho_\psi \psi_{ij,t-1} + \sigma_\psi \varepsilon_{ij,t}^\psi, \quad \varepsilon_{ij,t}^\psi \sim \mathcal{N}(0, 1). \quad (17)$$

Movements in $\psi_{ij,t}$ shift the demand for foreign assets independently of their fundamental payoff. In equilibrium, arbitrageurs absorb these changes in asset demand. The extent to which they do so, however, depends on the precision with which they can evaluate the return on the foreign asset.

3.6 Geographic Information Frictions

Arbitrageurs trade across countries based on the expected payoff from the international carry trade. Define the real payoff on the Foreign carry-trade position as

$$r_{t+1}^* \equiv \frac{\tilde{R}_{t+1}^*}{P_{t+1}^*}, \quad (18)$$

where

$$\tilde{R}_{t+1}^* = R_t^* - R_t \frac{E_t}{E_{t+1}}. \quad (19)$$

The arbitrageur does not observe the payoff perfectly. Instead, the observed return is

$$r_{t+1}^{\text{obs},*} = r_{t+1}^* + \nu_{t+1}, \quad (20)$$

where

$$\mathbb{E}_t[\nu_{t+1}] = 0, \quad \text{Var}_t(\nu_{t+1}) = \sigma_\nu^2 e^{\kappa \phi_{ij}}, \quad \phi_{ij} \equiv \ln \text{dist}_{ij}, \quad \kappa > 0. \quad (21)$$

Equation (21) is the key geographic friction in the model. Information about foreign returns becomes less precise as bilateral distance increases. Equivalently,

$$e^{\kappa \phi_{ij}} = \text{dist}_{ij}^\kappa,$$

so the variance of the information noise increases with geographic distance at elasticity κ .

Let the size of the arbitrage position be

$$x_t \equiv \frac{d_{t+1}^*}{R_t^*}. \quad (22)$$

Conditional on the arbitrageur's information set,

$$r_{t+1}^{\text{obs},*} \mid \mathcal{F}_t \sim \mathcal{N}(\mu_t, V(\phi_{ij})), \quad (23)$$

where

$$\mu_t = \mathbb{E}_t[r_{t+1}^*], \quad (24)$$

$$V(\phi_{ij}) = \text{Var}_t(r_{t+1}^*) + \sigma_\nu^2 e^{\kappa \phi_{ij}}. \quad (25)$$

Arbitrageurs have CARA preferences with risk-aversion parameter ω . Their portfolio problem is

$$\max_{x_t} \mathbb{E}_t \left[-\frac{1}{\omega} \exp \left(-\omega x_t r_{t+1}^{\text{obs},*} \right) \right]. \quad (26)$$

Given conditional normality, this problem is equivalent to

$$\max_{x_t} \left\{ \mu_t x_t - \frac{\omega}{2} V(\phi_{ij}) x_t^2 \right\}. \quad (27)$$

The optimal arbitrage position is therefore

$$\frac{d_{t+1}^*}{R_t^*} = \frac{\mathbb{E}_t[r_{t+1}^*]}{\omega V(\phi_{ij})}. \quad (28)$$

Equation (28) illustrates the central mechanism. Holding the expected return differential fixed, a larger information variance reduces the arbitrageur's position. Because information risk increases with bilateral distance, arbitrageurs respond less aggressively to a given return differential between geographically distant countries.

3.7 The UIP Wedge

Bond-market clearing requires

$$B_{t+1} + N_{t+1} + D_{t+1} = 0, \quad B_{t+1}^* + N_{t+1}^* + D_{t+1}^* = 0. \quad (29)$$

Combining arbitrageur demand with bond-market clearing yields a modified uncovered interest parity condition of the form

$$i_t - i_t^* - \mathbb{E}_t \Delta e_{t+1} = \chi_1(\phi_{ij}) \psi_t - \chi_2(\phi_{ij}) b_{t+1}, \quad (30)$$

where ψ_t denotes the financial demand shock and b_{t+1} denotes the international bond position. The coefficients $\chi_1(\phi)$ and $\chi_2(\phi)$ depend on the amount of information risk faced by international arbitrageurs.

In particular, the loading of the UIP wedge on the financial shock can be written as

$$\chi_1(\phi) = \underbrace{\frac{n\omega\sigma_e^2}{\beta m}}_{\chi_1^{IM}} + \underbrace{\frac{n\beta\omega\sigma_\nu^2}{m} e^{\kappa\phi}}_{\text{geographic information friction}}. \quad (31)$$

The first component is the financial-market term present without geographic information frictions, while the second arises from distance-dependent information risk.

Similarly,

$$\chi_2(\phi) = \underbrace{\frac{Y\omega\sigma_e^2}{m}}_{\chi_2^{IM}} + \underbrace{\frac{Y\beta^2\omega\sigma_\nu^2}{m} e^{\kappa\phi}}_{\text{geographic information friction}}. \quad (32)$$

The geographic implication follows immediately from Equation (31). Since $\kappa > 0$,

$$\frac{\partial \chi_1(\phi)}{\partial \phi} = \frac{n\beta\omega}{m} \kappa \sigma_\nu^2 e^{\kappa\phi} > 0. \quad (33)$$

Therefore, for a given realization of the financial shock ψ_t , the induced UIP wedge is larger between geographically more distant countries.

The mechanism operates through arbitrage capacity. Greater distance raises the information risk associated with a foreign position, causing arbitrageurs to absorb less of a given noise-trader demand shock. A larger return differential is therefore required to clear international asset markets. As a result, identical financial shocks generate larger deviations from UIP across more distant country pairs.

3.8 Model Implications

The model contains two distinct geographic channels. The first operates through international financial markets. Geographic distance raises the information risk faced by arbitrageurs, limiting their willingness to absorb financial demand shocks. As shown above, this mechanism amplifies the UIP wedge generated by a given financial shock:

$$\frac{\partial \chi_1(\phi)}{\partial \phi} > 0.$$

Thus, otherwise identical financial shocks generate larger deviations from uncovered interest parity between geographically more distant countries.

This financial channel also affects international consumption risk sharing. A larger UIP wedge generates greater movements in exchange rates relative to consumption and weakens the relationship implied by complete international financial markets. Consequently, the model predicts that the Backus–Smith relationship deteriorates with bilateral distance. This provides a direct theoretical counterpart to the negative geographic gradient in the Backus–Smith correlation documented in Section ??.

The financial channel alone, however, does not generate all of the spatial patterns documented in the data. In particular, geographic segmentation in financial markets can produce a spatial gradient in the Backus–Smith relationship without necessarily generating the observed geographic pattern in the comovement between relative consumption and relative output.

To account for this second moment, the model also contains a real-side geographic channel through

iceberg trade costs. Since bilateral trade costs increase with geographic distance,

$$\ln \tau_{ij} = \alpha_\tau + \eta_\tau \ln \text{dist}_{ij},$$

real shocks are transmitted differently across nearby and distant country pairs. Distance-dependent trade costs therefore affect the relationship between relative consumption and relative output.

The two shocks also have different implications for consumption–output comovement. Financial shocks tend to move consumption and output in opposite directions, whereas productivity shocks generate positive consumption–output comovement. The observed risk-sharing moments therefore reflect both the composition of shocks and the geographic structure governing their transmission.

This distinction is important for the quantitative analysis. Financial shocks are needed to generate exchange-rate disconnect and departures from UIP, while productivity shocks provide the real-side force necessary to reproduce positive consumption–output comovement. Geographic information frictions determine how the financial component varies with bilateral distance, whereas geographic trade costs determine how the transmission of real shocks varies across country pairs.

Taken together, the model therefore predicts that geography should appear in several international risk-sharing moments, but through different economic channels. Geographic information frictions amplify financial-market disconnect with distance, while trade costs shape the spatial transmission of real shocks. The interaction of these two channels determines the joint behavior of exchange rates, consumption, and output across country pairs.

The quantitative analysis below asks whether a single parameterization of these mechanisms can reproduce the geographic gradients observed in the data. In particular, I evaluate the model’s implications for the Backus–Smith correlation, the volatility of exchange rates relative to consumption and output, deviations from UIP, and the Fama coefficient as bilateral distance increases.

3.9 Model Implications

The model contains two distinct geographic channels. The first operates through international financial markets. Geographic distance raises the information risk faced by arbitrageurs, limiting their willingness to absorb financial demand shocks. As shown above, this mechanism amplifies the

UIP wedge generated by a given financial shock:

$$\frac{\partial \chi_1(\phi)}{\partial \phi} > 0.$$

Thus, otherwise identical financial shocks generate larger deviations from uncovered interest parity between geographically more distant countries.

This financial channel also affects international consumption risk sharing. A larger UIP wedge generates greater movements in exchange rates relative to consumption and weakens the relationship implied by complete international financial markets. Consequently, the model predicts that the Backus–Smith relationship deteriorates with bilateral distance. This provides a theoretical counterpart to the negative geographic gradient in the Backus–Smith correlation documented in the empirical analysis.

The financial channel alone, however, does not generate all of the spatial patterns documented in the data. In particular, geographic segmentation in financial markets can produce a spatial gradient in the Backus–Smith relationship without necessarily generating the observed geographic pattern in the comovement between relative consumption and relative output. To account for this second moment, the model also contains a real-side geographic channel through iceberg trade costs. Since bilateral trade costs increase with geographic distance,

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real shocks are transmitted differently across nearby and distant country pairs.

The two shocks also have different implications for consumption–output comovement. Financial shocks tend to generate negative consumption–output comovement, whereas productivity shocks generate positive comovement. The observed risk-sharing moments therefore reflect both the composition of shocks and the geographic structure governing their transmission. Financial shocks generate exchange-rate disconnect, while productivity shocks provide the real-side force necessary to reproduce positive consumption–output comovement. Geographic information frictions determine how the financial component varies with bilateral distance, whereas geographic trade costs shape the spatial transmission of real shocks.

Taken together, the model predicts that geography should appear in several international risk-sharing moments through distinct economic channels. The quantitative analysis below asks whether a single

parameterization of these mechanisms can reproduce the geographic gradients observed in the data. In particular, I evaluate the model’s implications for the Backus–Smith correlation, the volatility of exchange rates relative to consumption and output, deviations from UIP, and the Fama coefficient as bilateral distance increases.

4 Quantitative Analysis

I next evaluate whether the geographic mechanisms in the model can quantitatively reproduce the spatial patterns documented in the data. The exercise varies bilateral geographic distance while holding the remaining structural parameters fixed. Distance affects the economy through both channels introduced above: iceberg trade costs and information risk in international financial markets.

For each bilateral distance, I solve the model and compute a set of moments describing international risk sharing and exchange-rate behavior. I then compare how these moments change between relatively close and distant country pairs with their empirical counterparts. The quantitative exercise focuses on the Backus–Smith correlation, exchange-rate volatility relative to consumption and output, deviations from uncovered interest parity, and the Fama coefficient.

4.1 Parameterization

Table 1 summarizes the parameterization used in the quantitative exercise. The discount factor is set to $\beta = 0.99$ and the coefficient of relative risk aversion to $\sigma = 2$. The import-share parameter is $\gamma = 0.07$, while the elasticity of substitution between Home and Foreign goods is set to $\theta = 1.5$.

Two parameters govern how geography enters the model. The distance elasticity of iceberg trade costs is set to

$$\eta_\tau = 0.225,$$

so that bilateral trade costs increase systematically with geographic distance. The elasticity of information risk with respect to distance is

$$\kappa = 0.25.$$

The information-risk share at a bilateral distance of 1,000 km is set to $s_{\nu,0} = 0.25$.

Table 1: Parameterization

Parameter	Value	Description
β	0.99	Discount factor
σ	2	Risk aversion
γ	0.07	Import share parameter
θ	1.5	Trade elasticity
η_τ	0.225	Distance elasticity of trade costs
κ	0.25	Distance elasticity of information risk
$s_{\nu,0}$	0.25	Information-risk share at 1,000 km
ρ_a	0.97	Productivity-shock persistence
ρ_ψ	0.97	Financial-shock persistence
σ_a	10	Productivity-shock size
σ_ψ	1	Financial-shock size

Both productivity and financial shocks are persistent. Their autoregressive coefficients are set to

$$\rho_a = \rho_\psi = 0.97.$$

The innovation scales are $\sigma_a = 10$ for productivity shocks and $\sigma_\psi = 1$ for financial shocks.

4.2 Geographic Distance Experiment

To isolate the quantitative role of geography, I vary bilateral distance from 500 km to 12,000 km.

All structural parameters other than the geographic objects implied by distance are held fixed.

An increase in distance affects the economy through two margins. First, bilateral trade costs increase according to

$$\ln \tau(d) = \alpha_\tau + \eta_\tau \ln d.$$

Second, the variance of the information noise faced by international arbitrageurs increases according to

$$\text{Var}(\nu \mid d) = \sigma_\nu^2 d^\kappa.$$

Thus, moving from a nearby to a distant country pair simultaneously reduces real-side integration through trade costs and financial integration through information frictions.

For each distance, I compute the stochastic equilibrium of the model and recover the corresponding international risk-sharing and exchange-rate moments. I compare the model-implied change between 500 km and 12,000 km with the geographic gradients observed in the data.

4.3 Model versus Data

Table 2 compares the geographic changes in the model moments with their empirical counterparts. The model reproduces the central qualitative feature of the data: international risk sharing deteriorates and exchange-rate disconnect becomes stronger as bilateral geographic distance increases.

Backus–Smith correlation. The model generates a declining Backus–Smith correlation with geographic distance. As bilateral distance increases from 500 km to 12,000 km, the correlation between the real exchange rate and relative consumption growth falls from 0.038 to -0.018 . The corresponding empirical correlation falls from approximately 0.03 for nearby country pairs to approximately -0.06 for distant pairs.

Thus, the model reproduces both the direction of the geographic gradient and the transition from a weakly positive relationship among nearby countries to a negative relationship among distant countries. The result reflects the amplification of financial-market disconnect induced by distance-dependent information risk.

Relative exchange-rate volatility. Geographic information frictions also increase exchange-rate volatility relative to real quantities. The model-implied ratio

$$\frac{SD(\Delta e)}{SD(\Delta c^{rel})}$$

risks from 2.27 at 500 km to 3.10 at 12,000 km. Across distance, the implied slope is 0.26, compared with an empirical slope of 0.23.

A similar pattern appears when exchange-rate volatility is normalized by relative output growth.

The model-implied ratio

$$\frac{SD(\Delta e)}{SD(\Delta y^{rel})}$$

increases from 1.52 to 2.31, generating a geographic slope of 0.25. In the data, the corresponding ratio rises from approximately 2.6 to 3.5, with a slope of 0.28.

Although the model does not exactly match the level of every volatility ratio, it closely reproduces how relative exchange-rate volatility changes with geographic distance. This distinction is important for the exercise: the primary target is the cross-sectional geographic gradient rather than the unconditional level of exchange-rate volatility.

Table 2: Geographic Distance: Model versus Data

Moment	Model: 500–12,000 km	Data
Backus–Smith correlation	$0.038 \rightarrow -0.018$	$0.03 \rightarrow -0.06$
$SD(\Delta e)/SD(\Delta c^{rel})$	$2.27 \rightarrow 3.10$; slope 0.26	mean 2.57; slope 0.23
$SD(\Delta e)/SD(\Delta y^{rel})$	$1.52 \rightarrow 2.31$; slope 0.25	$2.6 \rightarrow 3.5$; slope 0.28
SD(UIP deviation)	$16.2 \rightarrow 23.0$ (+42%)	$7.2 \rightarrow 10.6\text{--}11$ (+50%)
Fama β	$-0.35 \rightarrow -0.51$	$0.16 \rightarrow -0.49$

UIP deviations. The model also produces larger deviations from uncovered interest parity for more distant country pairs. The standard deviation of the UIP deviation increases from 16.2 to 23.0 as distance rises from 500 km to 12,000 km, an increase of approximately 42 percent. In the data, the corresponding measure increases from approximately 7.2 percent to 10.6–11 percent, or roughly 50 percent.

The model therefore closely reproduces the proportional increase in UIP deviations with bilateral distance. This result corresponds directly to the analytical comparative static derived in Section 3.9: geographic information risk increases the loading of the UIP wedge on financial shocks.

Fama coefficient. Finally, the model generates a declining Fama coefficient as geographic distance increases. The coefficient falls from -0.35 at 500 km to -0.51 at 12,000 km. The empirical counterpart declines from approximately 0.16 among nearby country pairs to -0.49 among distant pairs.

The model therefore captures the increasingly negative Fama relationship among geographically distant countries. The quantitative mechanism is the same one underlying the UIP result: greater information risk weakens arbitrage across distant markets, allowing financial demand shocks to generate larger exchange-rate responses relative to interest-rate differentials.

Overall, the quantitative exercise shows that the model can reproduce several distinct geographic patterns in international macroeconomic data with a common mechanism. As bilateral distance increases, the model generates weaker Backus–Smith risk sharing, greater exchange-rate volatility relative to real quantities, larger UIP deviations, and a more negative Fama coefficient.

The model is particularly successful in reproducing the geographic gradients in relative exchange-rate volatility and the proportional increase in UIP deviations. The fit is not exact in levels for every moment, especially for the Fama coefficient among nearby country pairs. Nevertheless, the common

movement of these moments with distance suggests that geographic information frictions provide a quantitatively relevant source of exchange-rate disconnect.

5 Conclusion

This paper studies whether exchange-rate disconnect has a geographic dimension. I first document that several measures of international financial integration vary systematically with bilateral distance. The Backus–Smith correlation declines with geographic distance, relative consumption becomes more closely tied to relative output among distant country pairs, and the Fama relationship deteriorates with distance. Taken together, these findings suggest that international risk sharing and exchange-rate dynamics are systematically related to geography.

To interpret these patterns, I develop a two-country open-economy model in which international financial intermediation is subject to geographic information frictions. Arbitrageurs observe foreign asset returns imperfectly, and the variance of this information noise increases with bilateral distance. Greater information risk reduces arbitrageurs’ willingness to absorb financial demand shocks. As a result, the UIP wedge generated by a given financial shock increases with geographic distance. The model therefore provides a mechanism through which identical financial disturbances can generate different exchange-rate responses across country pairs.

The model also incorporates distance-dependent trade costs, which provide a distinct real-side geographic channel. While information frictions govern the transmission of financial shocks through asset markets, trade costs shape the transmission of real shocks. Together, these two channels allow the model to account for spatial variation in exchange rates, consumption, and output.

Quantitatively, the model reproduces several geographic patterns observed in the data. As bilateral distance increases from 500 to 12,000 kilometers, the model generates a declining Backus–Smith correlation, increasing exchange-rate volatility relative to consumption and output, larger deviations from uncovered interest parity, and a more negative Fama coefficient. In particular, the model closely matches the geographic gradients in relative exchange-rate volatility and the proportional increase in UIP deviations with distance.

Overall, the results suggest that the degree of international financial integration is not uniform across space. Geographic distance affects not only trade in goods but also the capacity of financial

markets to intermediate shocks. Incorporating this spatial dimension into models of exchange-rate determination can therefore help account for systematic heterogeneity in exchange-rate disconnect across countries.

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